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import time
import Adafruit_DHT
import spidev
import serial
from RPLCD.i2c import CharLCD

# Initialize DHT11 sensor
DHT_SENSOR = Adafruit_DHT.DHT11
DHT_PIN = 4 # GPIO4

# Initialize SPI for MCP3008
spi = spidev.SpiDev()
spi.open(0, 0)
spi.max_speed_hz = 1350000

# Initialize I2C LCD
lcd = CharLCD('PCF8574', 0x27)

# Initialize serial for GSM module
gsm_serial = serial.Serial("/dev/serial0", baudrate=9600, timeout=1)

# Thresholds
TEMP_THRESHOLD = 35.0
PULSE_THRESHOLD = 40

def read_adc(channel):
    adc = spi.xfer2([1, (8 + channel) << 4, 0])
    data = ((adc[1] & 3) << 8) + adc[2]
    return data

def send_sms(message):
    gsm_serial.write(b'AT+CMGF=1\r')
    time.sleep(1)
    gsm_serial.write(b'AT+CMGS="+1234567890"\r') # Replace with your phone
number
    time.sleep(1)
    gsm_serial.write(message.encode() + b"\r")
    time.sleep(1)
    gsm_serial.write(bytes([26]))
    time.sleep(3)

while True:
    humidity, temperature = Adafruit_DHT.read(DHT_SENSOR, DHT_PIN)

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pulse_value = read_adc(0) # Read from channel 0
pulse = int((pulse_value / 1023.0) * 100) # Simplified conversion
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lcd.clear()
if temperature is not None:
    lcd.write_string(f"Temp: {temperature:.1f}C")
else:
    lcd.write_string("Temp: Error")
lcd.crlf()
lcd.write_string(f"Pulse: {pulse} bpm")
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if temperature is not None and (temperature < TEMP_THRESHOLD or pulse <
PULSE_THRESHOLD):
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    alert_msg = "ALERT: "
    if temperature < TEMP_THRESHOLD:
        alert_msg += f"Low Temp ({temperature:.1f}C) "
    if pulse < PULSE_THRESHOLD:
        alert_msg += f"Low Pulse ({pulse} bpm)"
    send_sms(alert_msg)
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time.sleep(2)
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